Faster **prediction** of **wildfire** behaviour by physical models through application of proper orthogonal decomposition

E Guelpa, A Sciacovelli, V Verda… - … Journal of **Wildland Fire**, 2016 - CSIRO Publishing

… Reliable and computationally feasible **prediction** of **wildland fire** behaviour (eg rate of **spread**
[… A literature **review** revealed no applications of POD to **wildland fire** modelling. This paper …

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… predict the **behavior** of the … **simulation**, **prediction**, and visualization of **wildfire behavior**.
One goal of WIFIRE is to provide the tools to predict a more accurate rate of a spreading **wildfire**. …

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Evaluating crown fire rate of **spread** predictions from physics-based models

CM Hoffman, J Canfield, RR Linn, W Mell, CH Sieg… - Fire Technology, 2016 - Springer

… a variety of **wildland fire behavior** questions, there is a need to assess **model** performance,
especially in terms of crown fire **behavior**, to identify potential **model** errors and establish …

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Estimating **wildfire behavior** and effects

FA Albini - 1976 - books.google.com

… those concerned with **wildland fire behavior** and effects. Although fire **behavior prediction** is
by … Here we **review** some **model** uses and indicate the type of output needed and the general …

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Next day **wildfire spread**: A machine learning dataset to predict **wildfire** spreading from remote-sensing data

F Huot, RL Hu, N Goyal, T Sankar… - IEEE Transactions on …, 2022 - ieeexplore.ieee.org

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Using cellular automata to simulate **wildfire propagation** and to assist in fire management

JG Freire, CC DaCamara - Natural hazards and earth system …, 2019 - nhess.copernicus.org

… As mentioned in the introduction, we apply a CA **model** to a large and well-documented
wildfire that occurred in July 2012 in the Tavira and São Brás de Alportel municipalities, located …

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Parameter flexible **wildfire prediction** using machine learning techniques: Forward and inverse modelling

S Cheng, Y Jin, SP Harrison, C Quilodrán-Casas… - Remote Sensing, 2022 - mdpi.com

… observations; (ii) the forward fire **prediction model** once combined with ML approaches; (iii) …
propagation using estimated/adjusted parameters. After ROM, we aim to learn the **dynamics** …

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Empirical-based models for predicting head-fire rate of **spread** in Australian fuel types

MG Cruz, JS Gould, ME Alexander, AL Sullivan… - Australian …, 2015 - Taylor & Francis

… of each **model** of head-fire rate of **spread** and their … -fire rate of **spread** described in this
review are listed in Table … In presenting the various **bushfire** rate of **spread** models used in …

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A two-dimensional reaction-advection-diffusion **model** of the **spread** of fire in wildlands

P Grasso, M Innocente - Advances in **Forest Fire** …, 2018 - pureportal.coventry.ac.uk

… The aim of this research is to **model** the **spread** of **forest fire** efficiently in order to support the design of innovative fire management and suppression technologies and strategies, while …

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Stochastic modelling of **wildfire spread**

S Masoudian - 2024 - search.proquest.com

… **review** - This chapter provides an overview of the mathematical modeling of **wildfire spread** … and another generated using the **Wildfire** Dynamic Simulator (wfds) **simulation**. The aim is to …

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